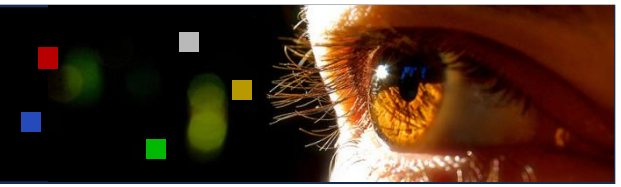
Visión por Computador

Versión 2026

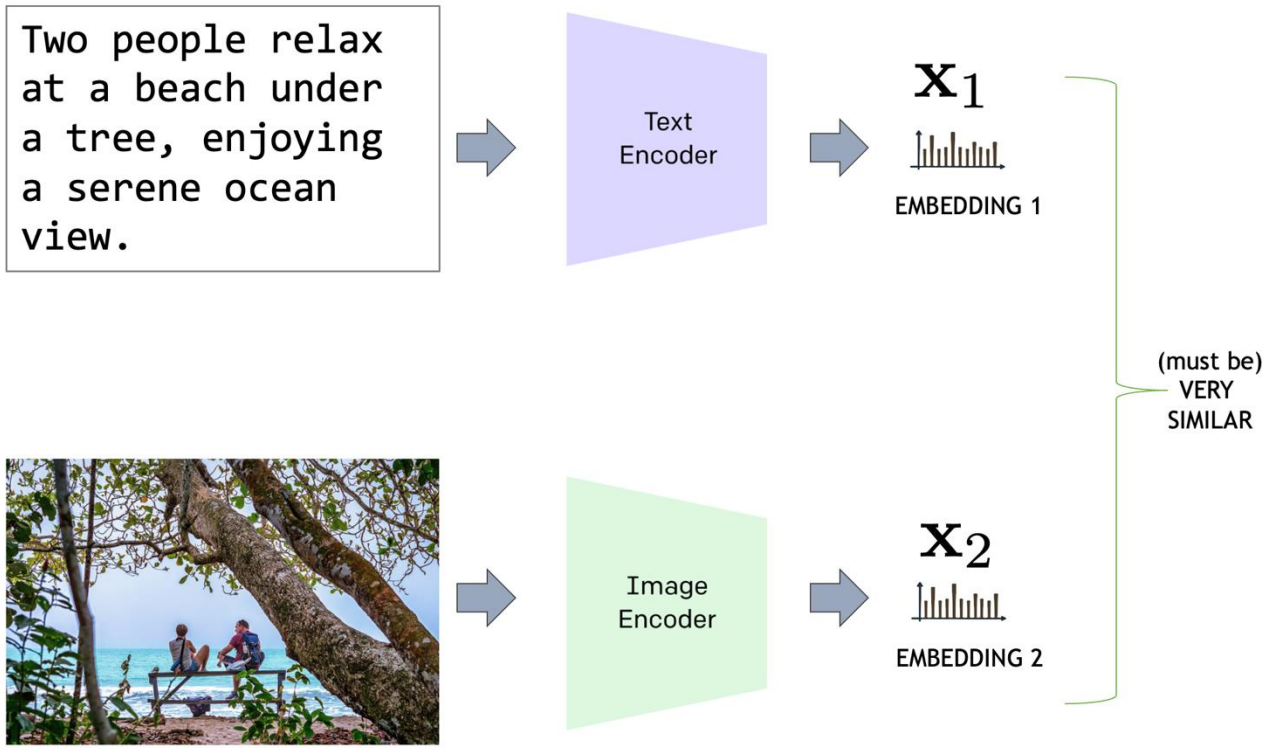


CLIP

(CAPÍTULO 3 – CV03G)

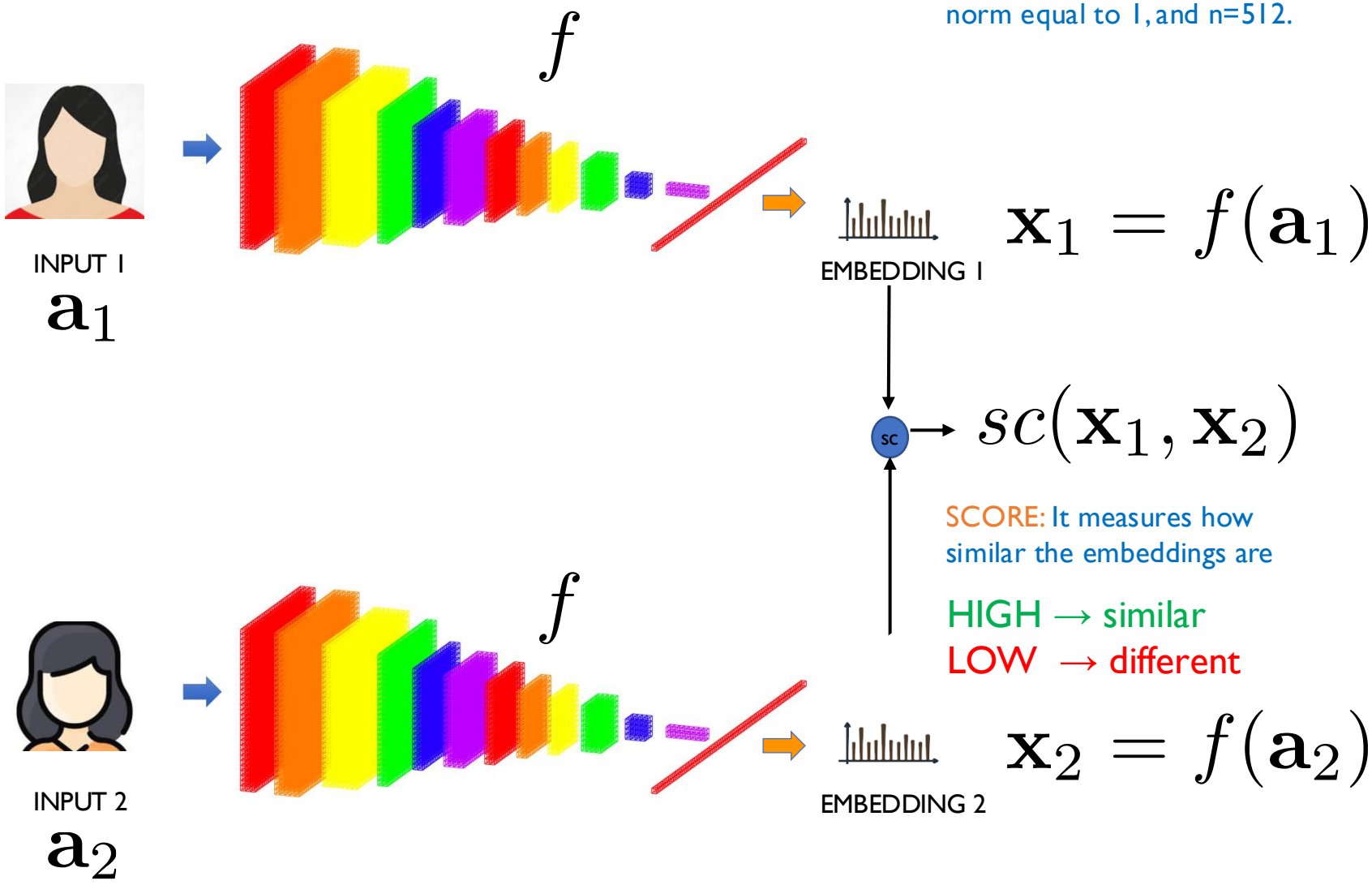
Domingo Mery

Ciencia de la Computación
Universidad Católica – Chile
http://domingomery.ing.uc.cl

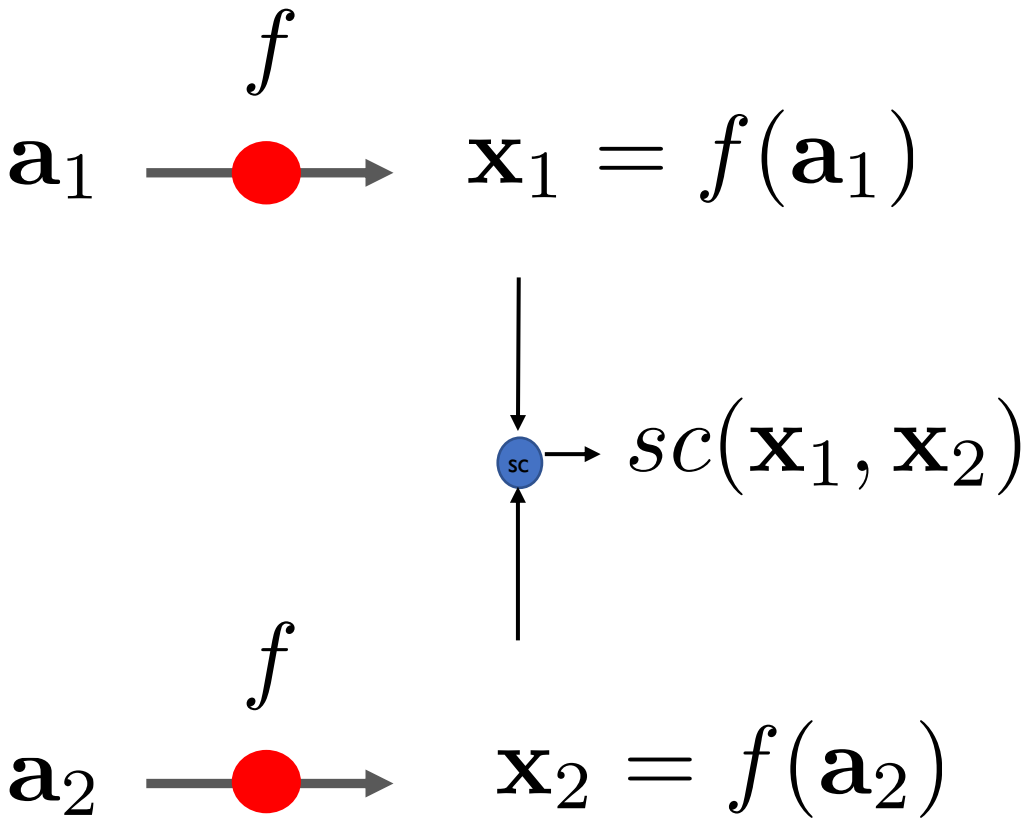


Embeddings

EMBEDDINGS: vectors of n elements that describe the input. Typically, they have a norm equal to 1, and n=512.



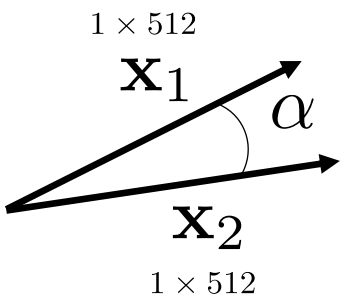
Embeddings


f
$\mathbf{a}_1$
$\mathbf{x}_1 = f(\mathbf{a}_1)$
sc
$sc(\mathbf{x}_1, \mathbf{x}_2)$
f
$\mathbf{a}_2$
$\mathbf{x}_2 = f(\mathbf{a}_2)$

Embeddings

HOW SIMILAR ARE TWO VECTORS?

DON'T FORGET: in our case the vectors have 512 elements and have a norm equal to 1



DOT PRODUCT

$$\langle \mathbf{x}_1, \mathbf{x}_2 \rangle = ||\mathbf{x}_1|| ||\mathbf{x}_2|| \cos(\alpha)$$

$$\langle \mathbf{x}_1, \mathbf{x}_2 \rangle = \cos(\alpha) \quad \text{SCORE}$$

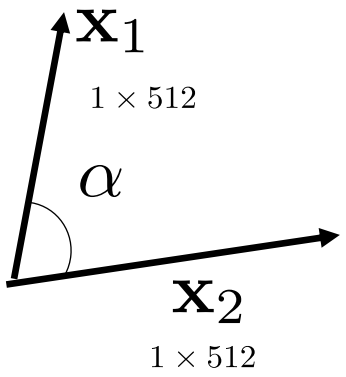
$$\langle \mathbf{x}_1, \mathbf{x}_2 \rangle = \mathbf{x}_1 \mathbf{x}_2^T$$

- If the vectors are very similar, the angle between them is small and the SCORE is close to 1.

Embeddings

HOW SIMILAR ARE TWO VECTORS?

DON'T FORGET: in our case the vectors have 512 elements and have a norm equal to 1



DOT PRODUCT

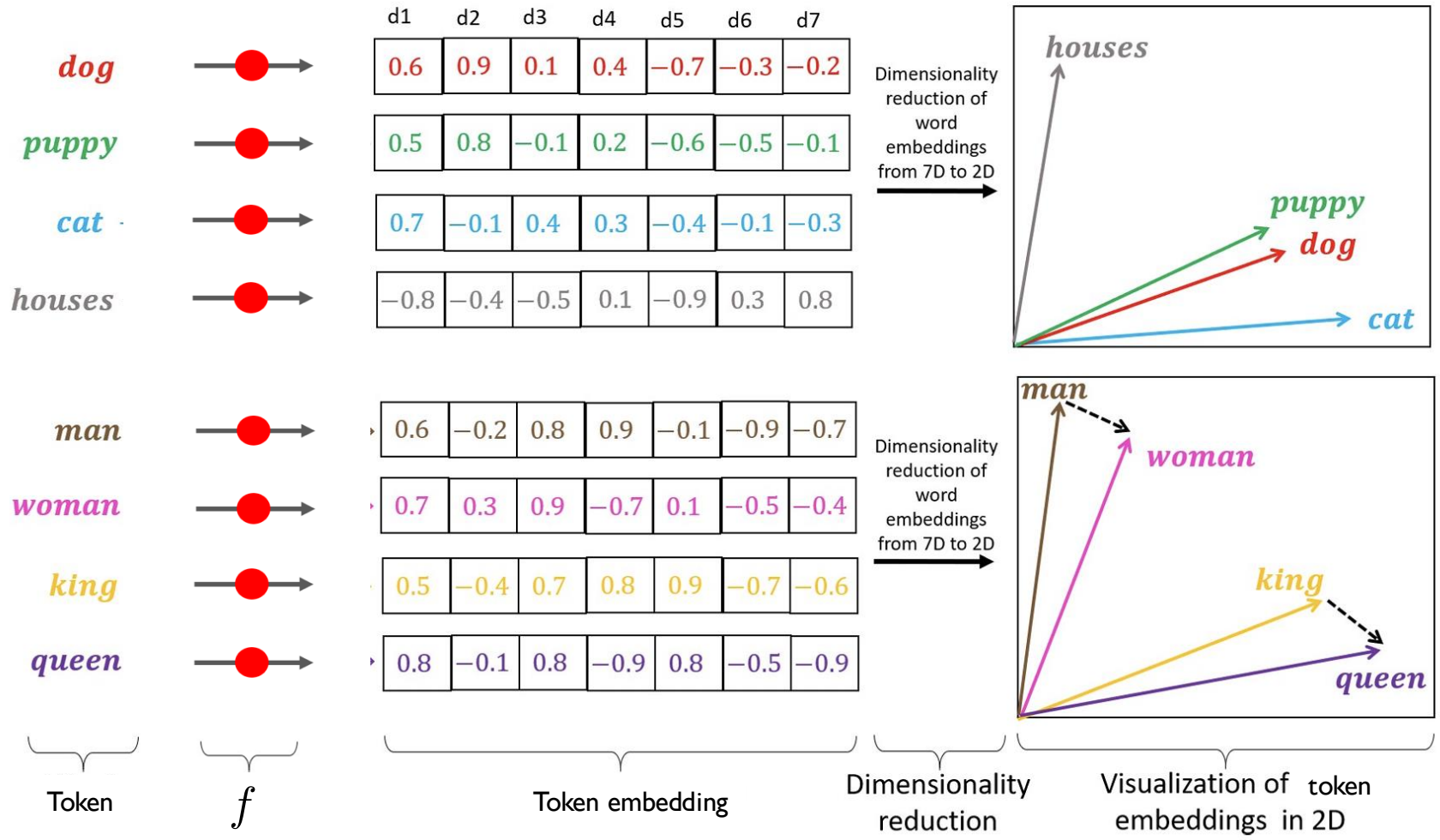
$$\langle \mathbf{x}_1, \mathbf{x}_2 \rangle = ||\mathbf{x}_1|| ||\mathbf{x}_2|| \cos(\alpha)$$

$$\langle \mathbf{x}_1, \mathbf{x}_2 \rangle = \cos(\alpha) \quad \text{SCORE}$$

$$\langle \mathbf{x}_1, \mathbf{x}_2 \rangle = \mathbf{x}_1 \mathbf{x}_2^T = sc(\mathbf{x}_1, \mathbf{x}_2)$$

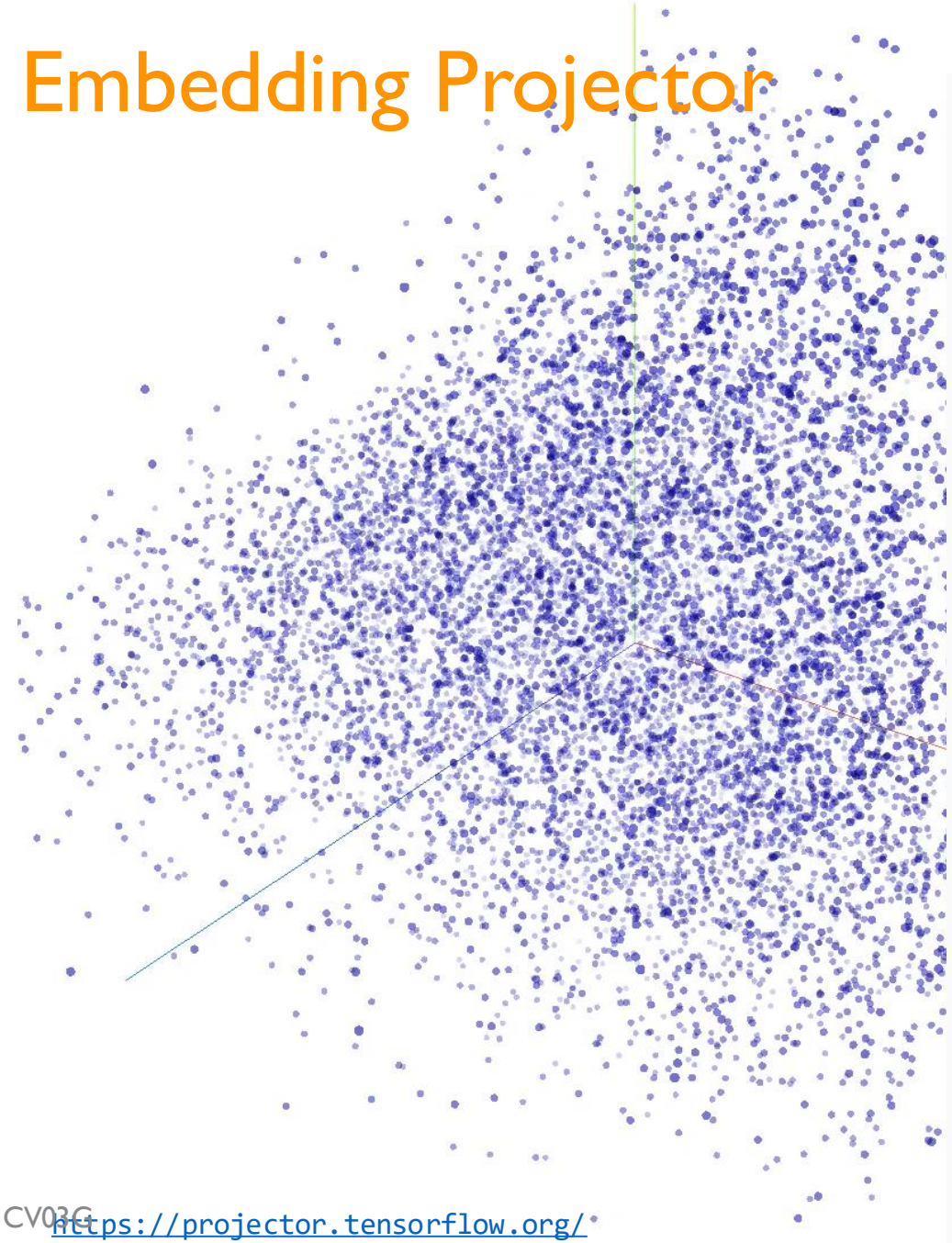
- If the vectors are very similar, the angle between them is small and the SCORE is close to 1.
- If the vectors are very different, the angle between them is large and the SCORE decreases significantly.

Embeddings in NLP



Adapted from: Rozado, D. (2020). Wide range screening of algorithmic bias in word embedding models using large sentiment lexicons reveals underreported bias types. PloS one, 15(4), e0231189.

Embedding Projector



Search travelled .* by word

neighbors 100

distance COSINE EUCLIDEAN

Nearest points in the original space:

traveled	0.344
travelling	0.492
went	0.506
visited	0.532
travels	0.541
journey	0.575
returned	0.578
moved	0.588
returning	0.591
worked	0.597
emigrated	0.607
travel	0.610
lived	0.611
arrived	0.613

https://projector.tensorflow.org/

Key Idea of CLIP

Two people relax at a beach under a tree, enjoying a serene ocean view.



Text Encoder

Image Encoder

$\mathbf{x}_1$

EMBEDDING 1

$\mathbf{x}_2$

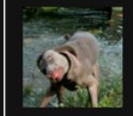
EMBEDDING 2

$$sc(\mathbf{x}_1, \mathbf{x}_2) = \mathbf{x}_1 \mathbf{x}_2^T \rightarrow \text{HIGH}$$

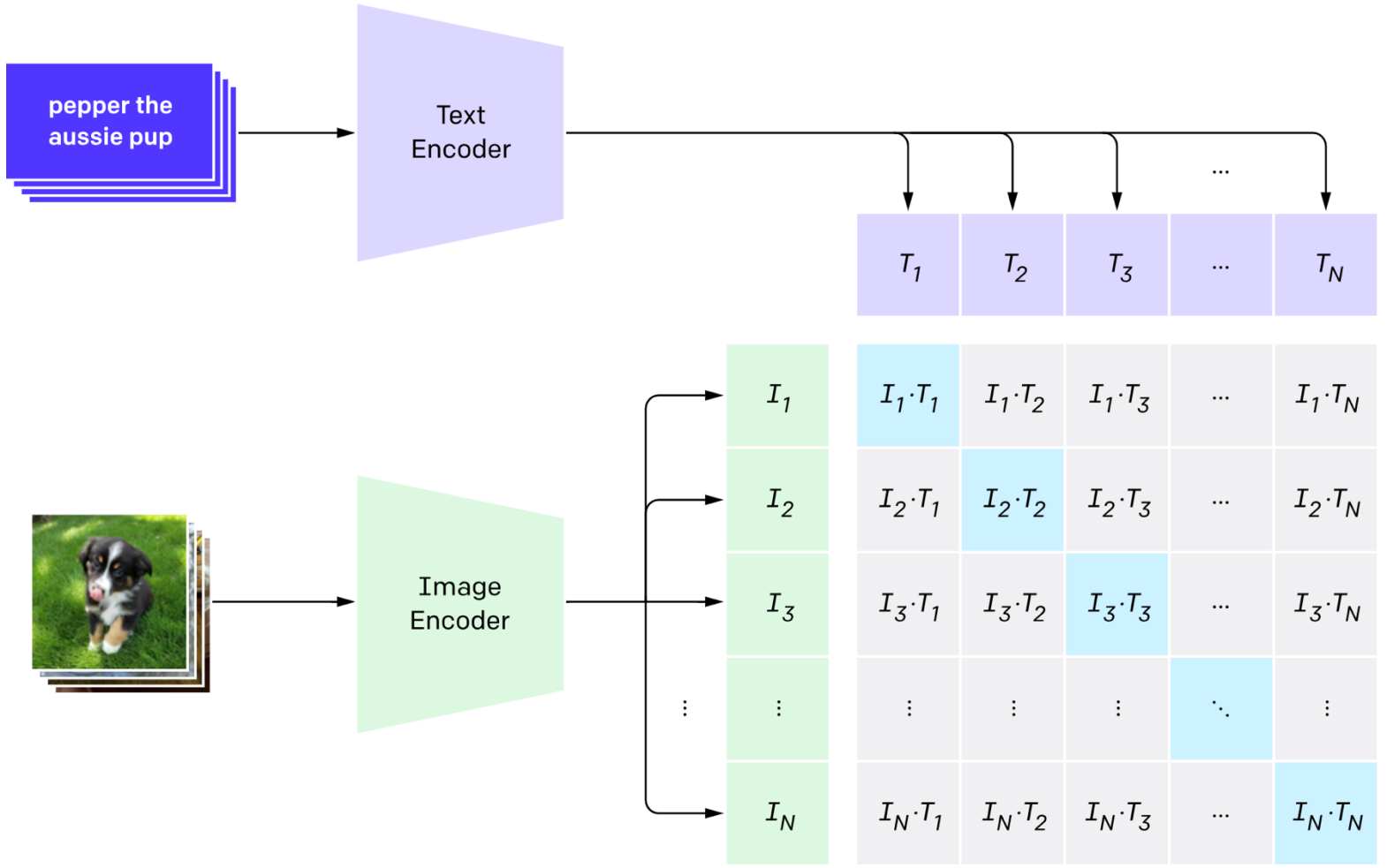
(must be) VERY SIMILAR

HOW TO TRAIN THEM?

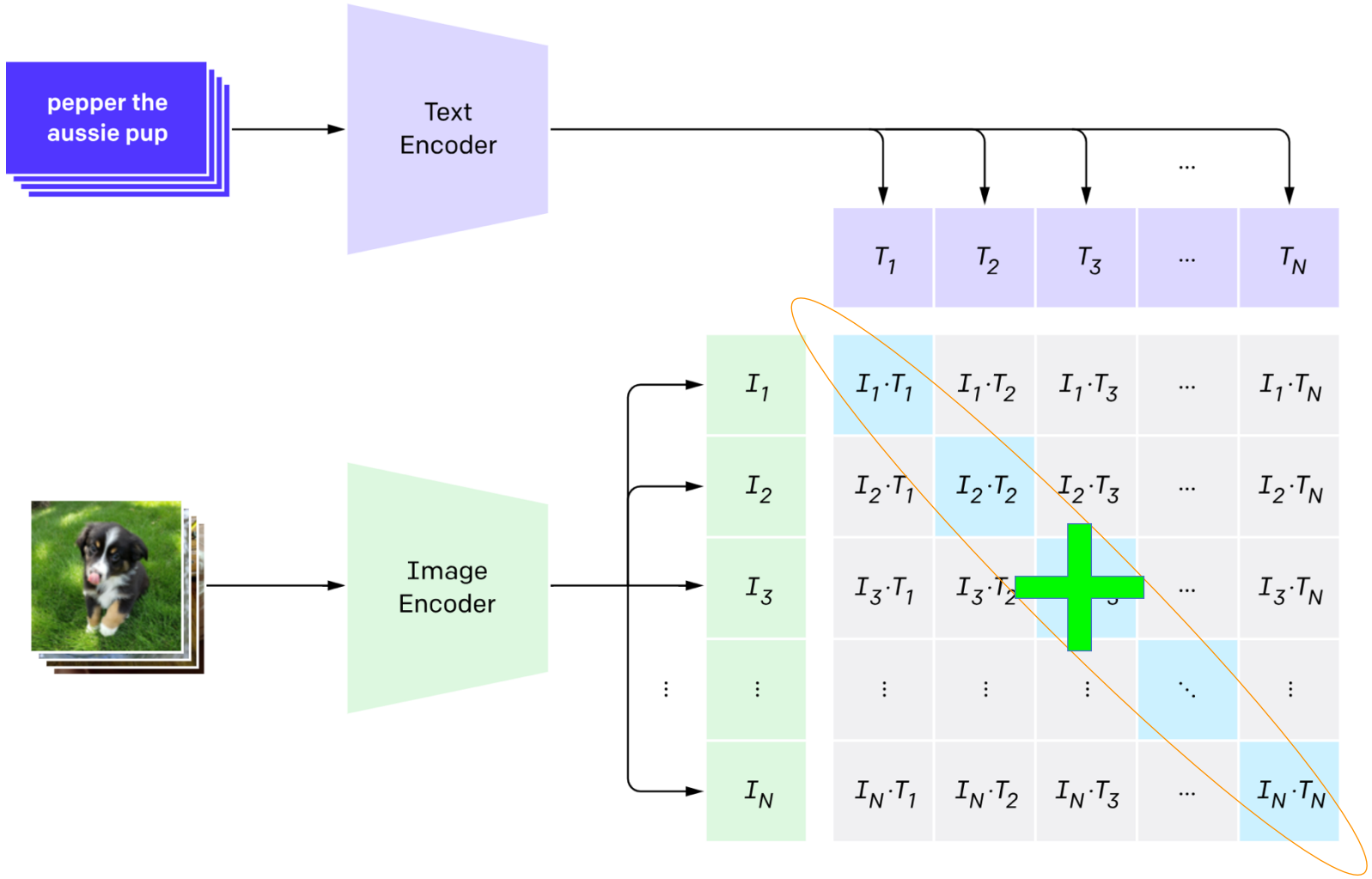
Data Collection: (Image , Text) 400M pairs

	A brown and white dog catches a Frisbee in it mouth in front of a group of people.		A pale dog runs down a path.
	A mostly black short haired dog is running in a swampy area with its tongue out.		Pictures on a white brick wall with a smiling man standing in front of the wall.
	A man wearing a red jacket is sitting on a wooden bench and is cooking something in a small pot.		Two people are in the water and splashing.
	A man stands on the rocky ledge of a large cliff.		A large black limousine parked in front of a building with a lot of people on the steps.
	The boy is performing a trick on a red rail with a skateboard.		The dog sits by a snowdrift.
	One climber is sitting on the rock overlooking the ocean whilst another climber is on her way up.		A girl in pink picks flowers.
	A brown dog shaking off water.		"A group of people walk in the park, while some talk on phones."
	A child on an amusement park swing ride.		The dog leaps into the air in a field.

Contrastive Pre-Training

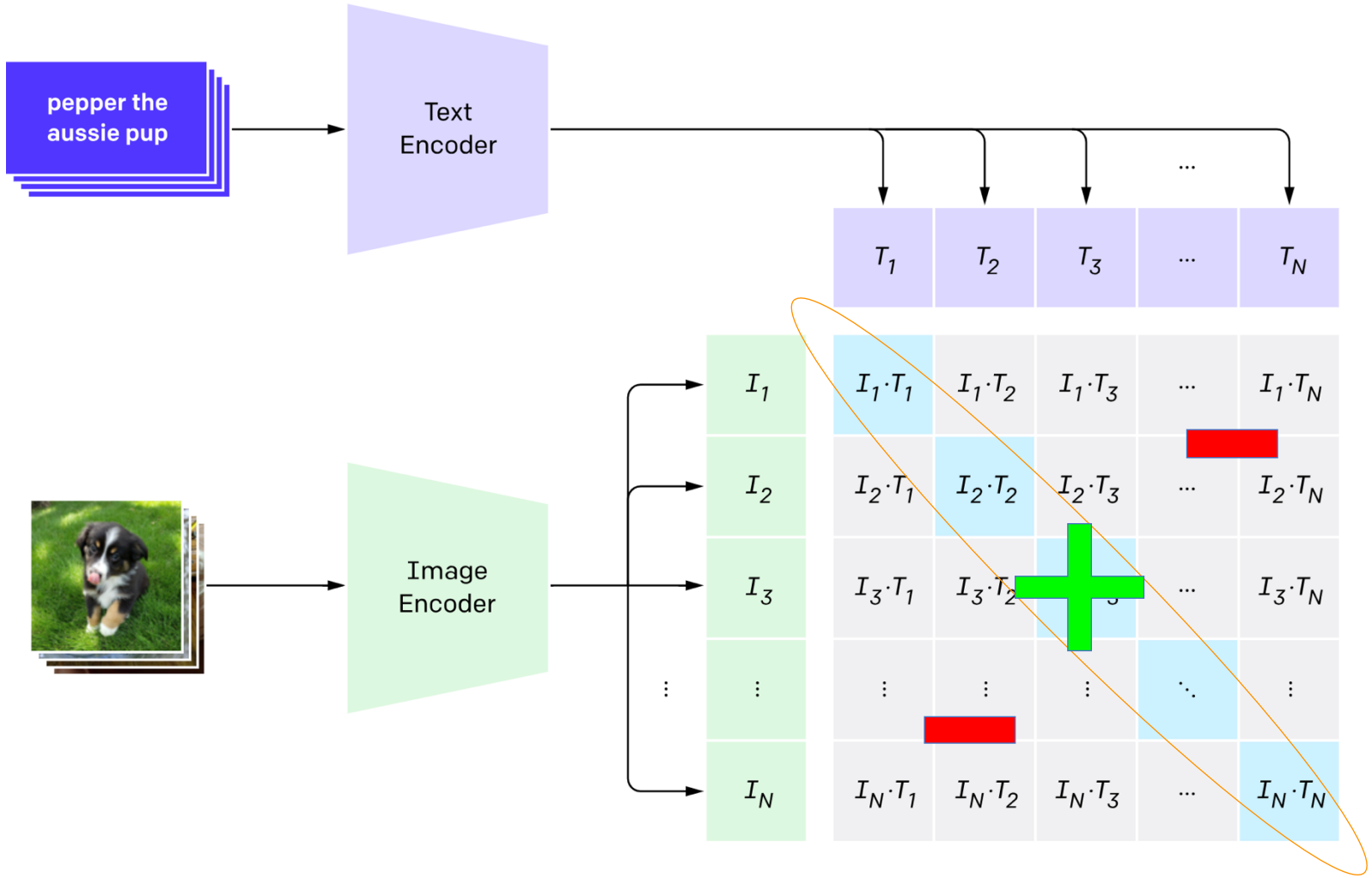


Contrastive Pre-Training



Maximize the diagonal

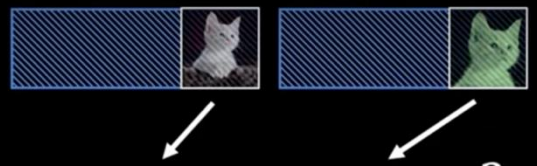
Contrastive Pre-Training



Contrastive loss: minimizes distance between correct pairs, maximizes for incorrect pairs.

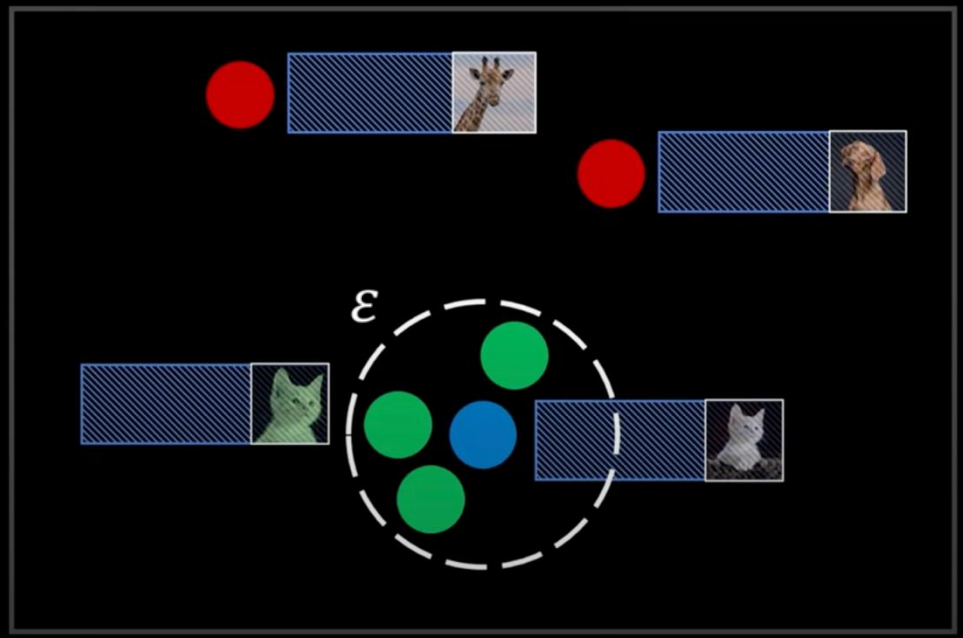
Maximize the diagonal,
and minimize the non-diagonal

Max Margin Loss



$$L(x_i, x_j) = \mathbb{1}[y_i = y_j]\|M(x_i) - M(x_j)\|_2^2 +$$
$$\mathbb{1}[y_i \neq y_j]\max(0, \varepsilon - \|M(x_i) - M(x_j)\|_2)^2$$





2005

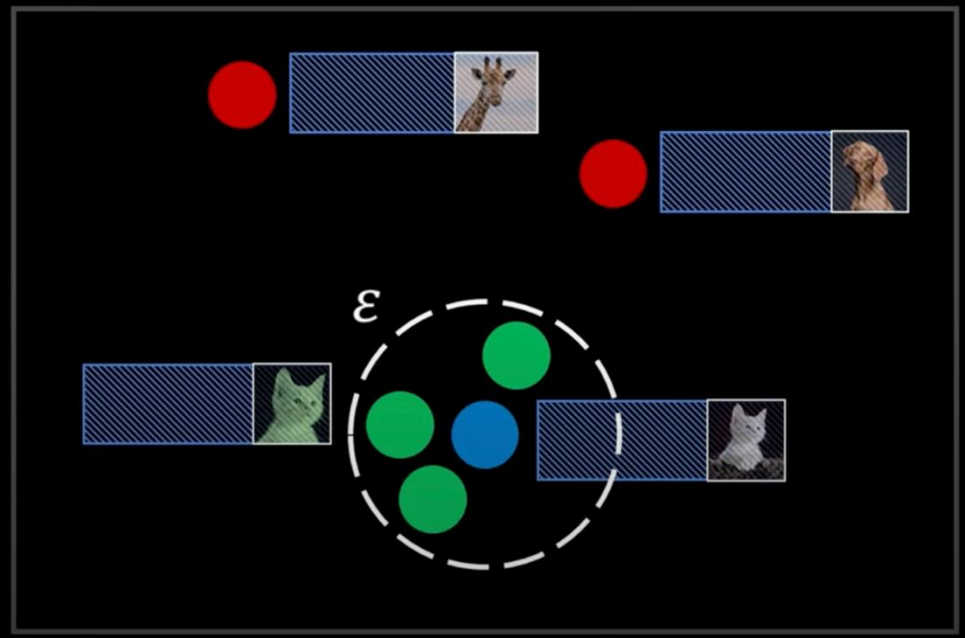
Sumit Chopra, Raia Hadsell, Yann LeCun

Max Margin Loss

$$L(x_i, x_j) = \mathbb{1}[y_i = y_j]\|M(x_i) - M(x_j)\|_2^2 + \mathbb{1}[y_i \neq y_j]\max(0, \varepsilon - \|M(x_i) - M(x_j)\|_2)^2$$



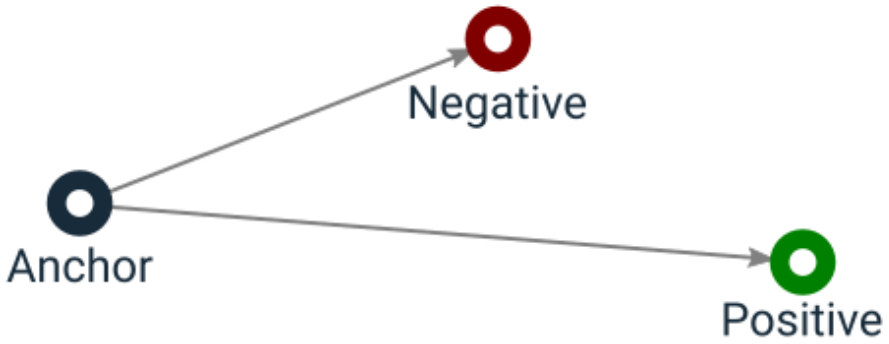




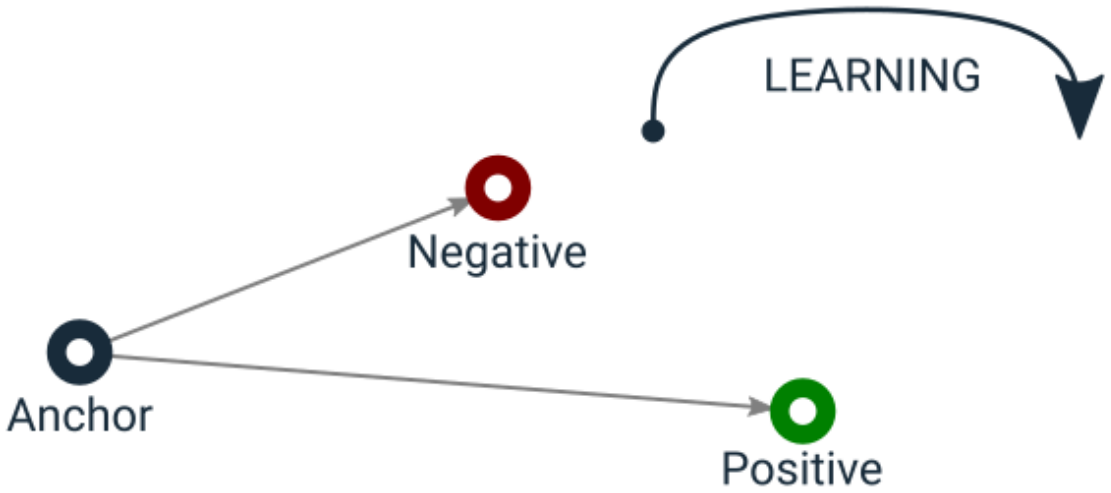
2005

Sumit Chopra, Raia Hadsell, Yann LeCun

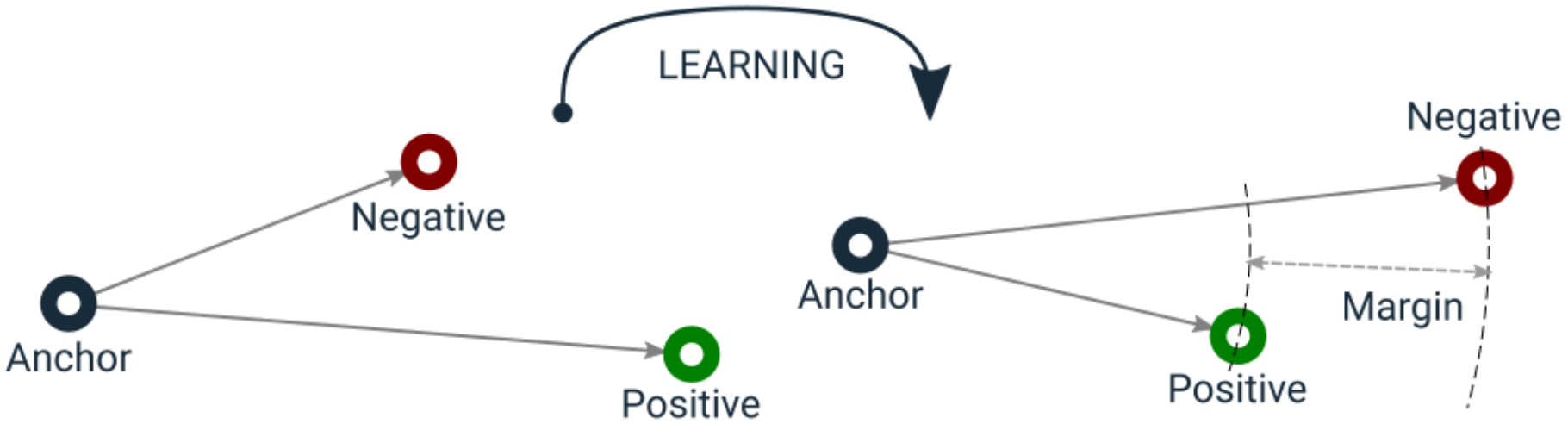
Triplet Loss



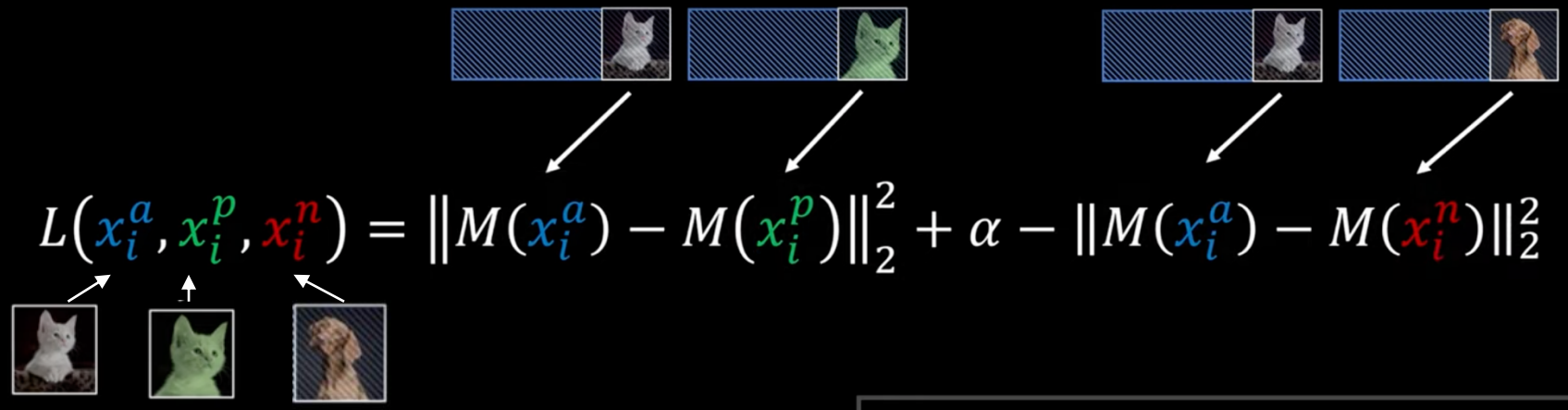
Triplet Loss



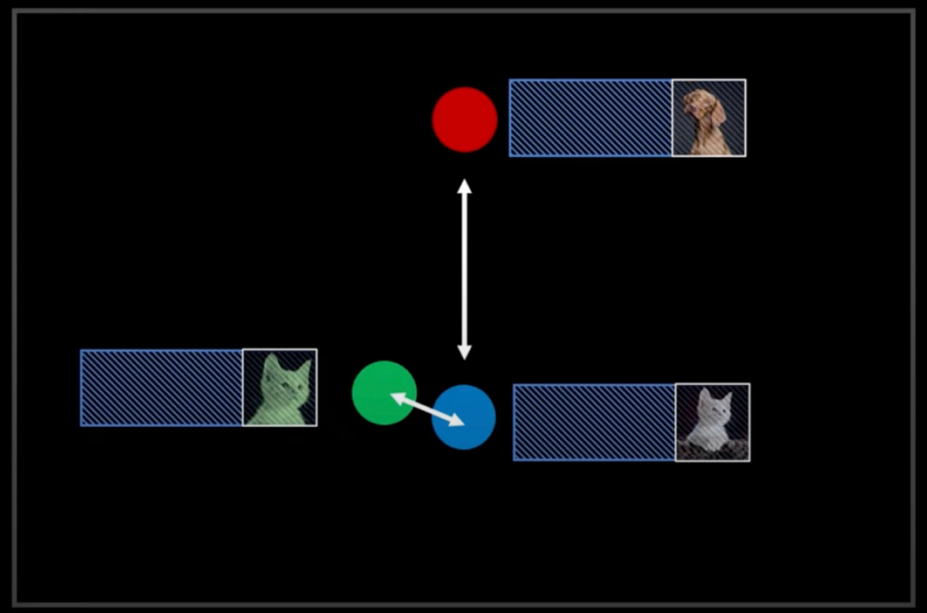
Triplet Loss



Triplet Loss



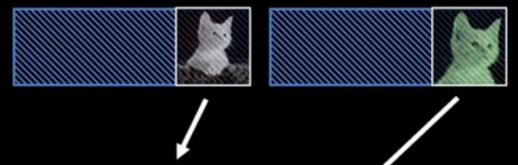
$$L(x_i^a, x_i^p, x_i^n) = \|M(x_i^a) - M(x_i^p)\|_2^2 + \alpha - \|M(x_i^a) - M(x_i^n)\|_2^2$$



2015

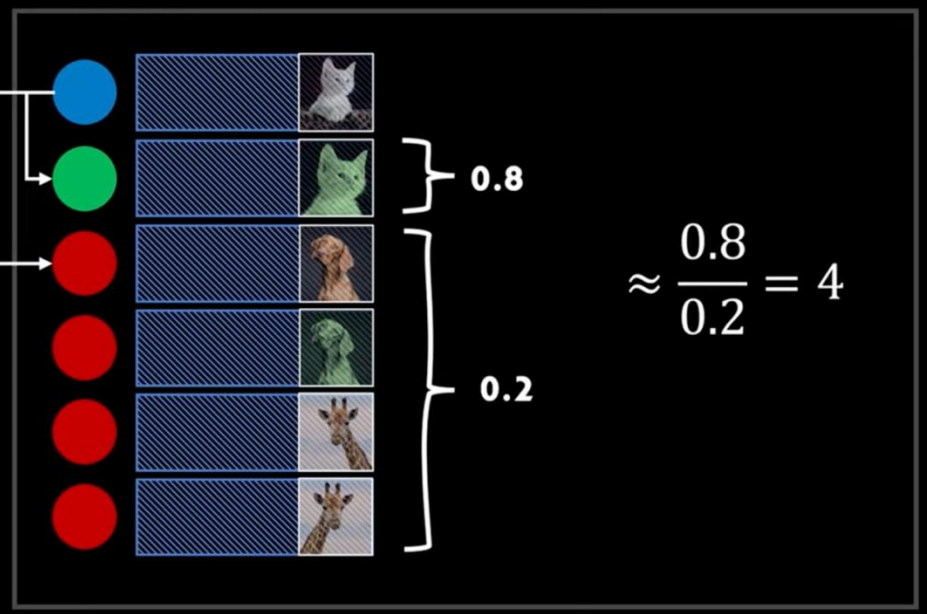
Florian Schroff, Dmitry Kalenichenko, James Philbin

NT-Xent Loss



$$L(x_i, x_j) = -\log \frac{\exp\left(sim\left(M(x_i), M(x_j)\right)\right)/\mathrm{T}}{\sum_{k=1}^{2N} \mathbb{1}[k \neq i] \exp\left(sim\left(M(x_i), M(x_k)\right)\right)/\mathrm{T}}$$



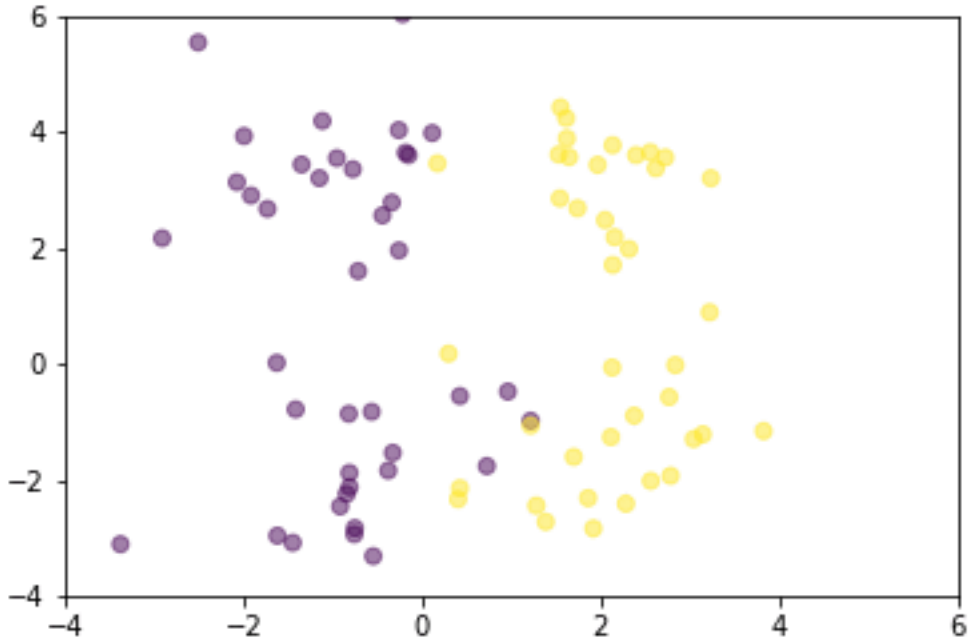


$$\approx \frac{0.8}{0.2} = 4$$

2020

Ting Chen, Simon Kornblith, Mohammad Norouzi, Geoffrey Hinton

Contrastive Loss



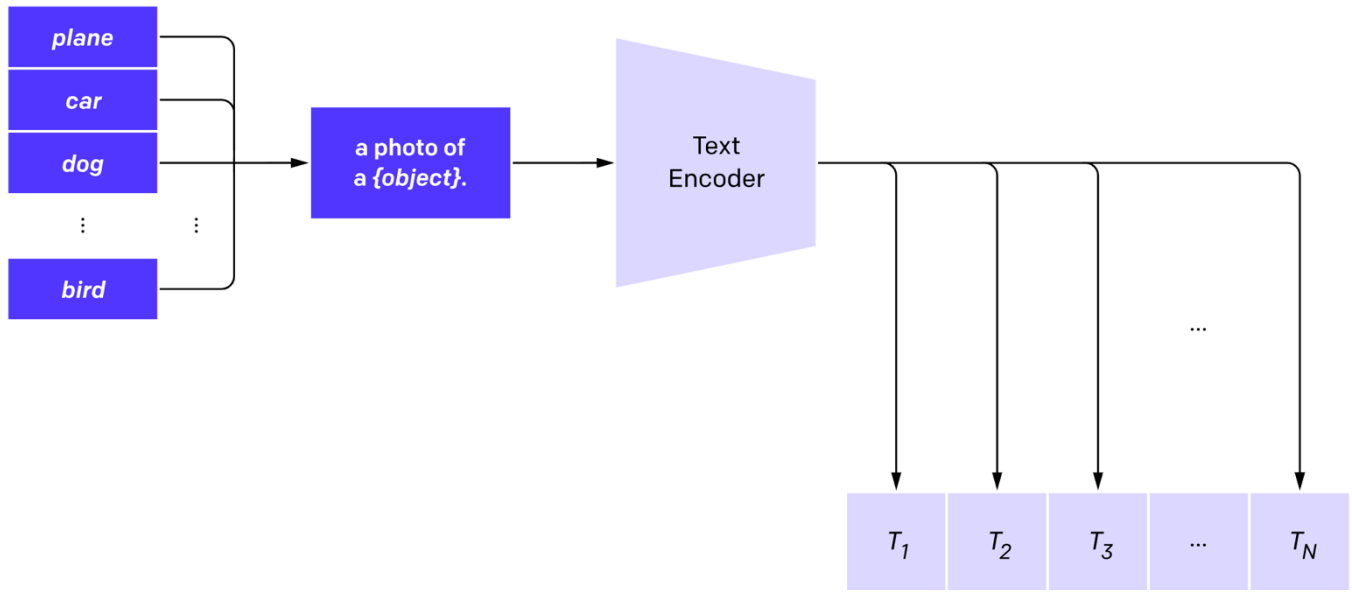
Contrastive Pre-Training



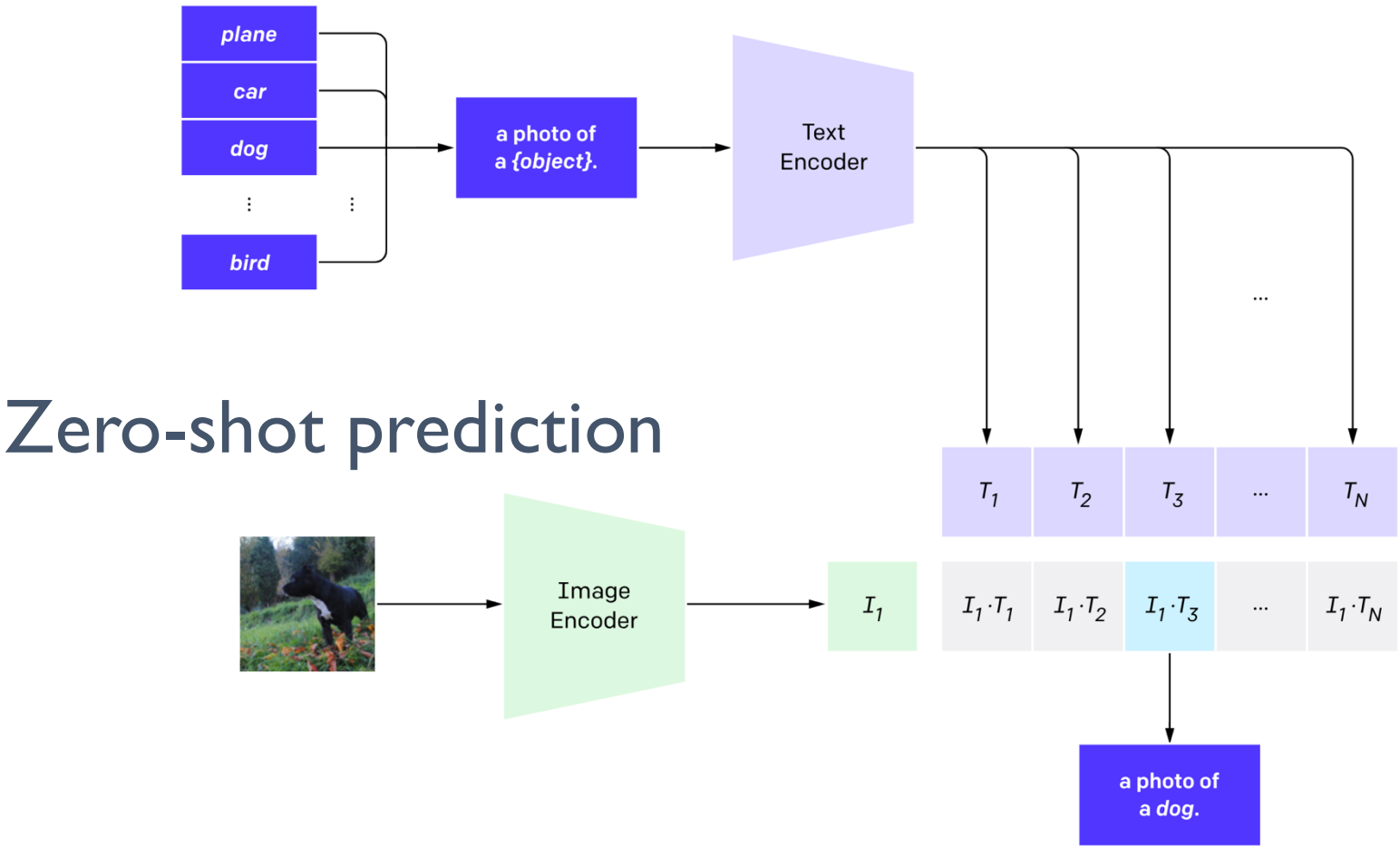
Applications: Zero-Shot Classification

CLIP can classify images into categories that were not included during its training. This is possible because it generates image and text embeddings that are directly comparable, enabling the model to predict categories based solely on textual descriptions provided at runtime.

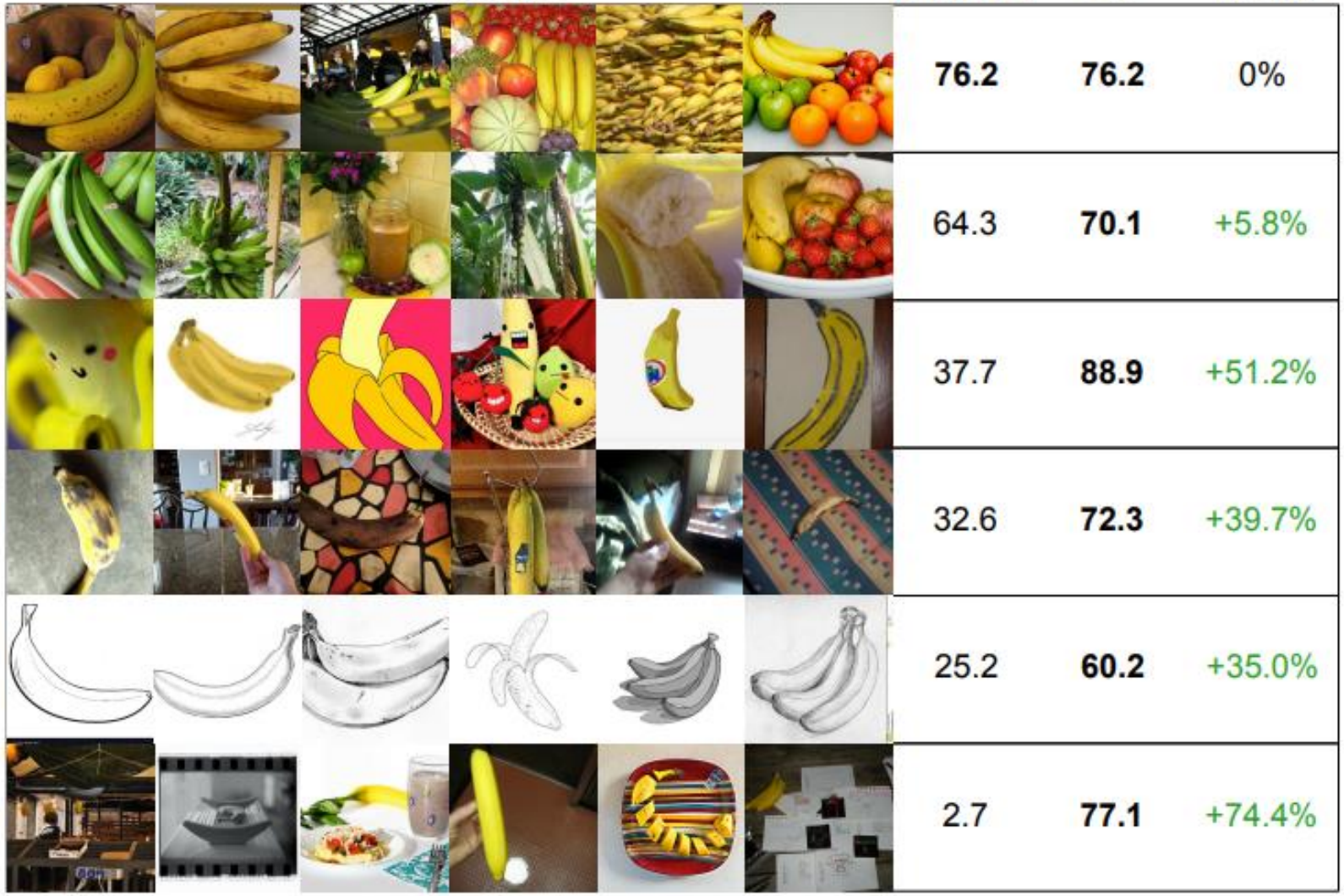
Create Dataset Classifier from label Text



Create Dataset Classifier from label Text



Comparison


Dataset Examples
ImageNet ResNet101
Zero-Shot CLIP
Δ Score
ImageNet
76.2
76.2
0%
ImageNetV2
64.3
70.1
+5.8%
ImageNet-R
37.7
88.9
+51.2%
ObjectNet
32.6
72.3
+39.7%
ImageNet Sketch
25.2
60.2
+35.0%
ImageNet-A
2.7
77.1
+74.4%

Results

Food101

guacamole (90.1%) Ranked 1 out of 101 labels



✓ a photo of **guacamole**, a type of food.

× a photo of **ceviche**, a type of food.

× a photo of **edamame**, a type of food.

× a photo of **tuna tartare**, a type of food.

× a photo of **hummus**, a type of food.

SUN397

television studio (90.2%) Ranked 1 out of 397 labels



✓ a photo of a **television studio**.

× a photo of a **podium indoor**.

× a photo of a **conference room**.

× a photo of a **lecture room**.

× a photo of a **control room**.

Youtube-BB

airplane, person (89.0%) Ranked 1 out of 23 labels



✓ a photo of a **airplane**.

× a photo of a **bird**.

× a photo of a **bear**.

× a photo of a **giraffe**.

× a photo of a **car**.

EuroSAT

annual crop land (46.5%) Ranked 4 out of 10 labels



× a centered satellite photo of **permanent crop land**.

× a centered satellite photo of **pasture land**.

× a centered satellite photo of **highway or road**.

✓ a centered satellite photo of **annual crop land**.

× a centered satellite photo of **brushland or shrubland**.

Results

Kinetics-700

country line dancing (99.0%) Ranked 1 out of 700 labels



- ✓ a photo of **country line dancing**.
- ✗ a photo of **square dancing**.
- ✗ a photo of **swing dancing**.
- ✗ a photo of **dancing charleston**.
- ✗ a photo of **salsa dancing**.

Flowers-102

great masterwort (74.3%) Ranked 1 out of 102 labels



- ✓ a photo of a **great masterwort**, a type of flower.
- ✗ a photo of a **bishop of llandaff**, a type of flower.
- ✗ a photo of a **pincushion flower**, a type of flower.
- ✗ a photo of a **globe flower**, a type of flower.
- ✗ a photo of a **prince of wales feathers**, a type of flower.

ImageNet

King Charles Spaniel (91.6%) Ranked 1 out of 1000 labels



- ✓ a photo of a **king charles spaniel**.
- ✗ a photo of a **brittany dog**.
- ✗ a photo of a **cocker spaniel**.
- ✗ a photo of a **papillon**.
- ✗ a photo of a **sussex spaniel**.

Birdsnap

Black chinned Hummingbird (33.3%) Ranked 4 out of 500 labels

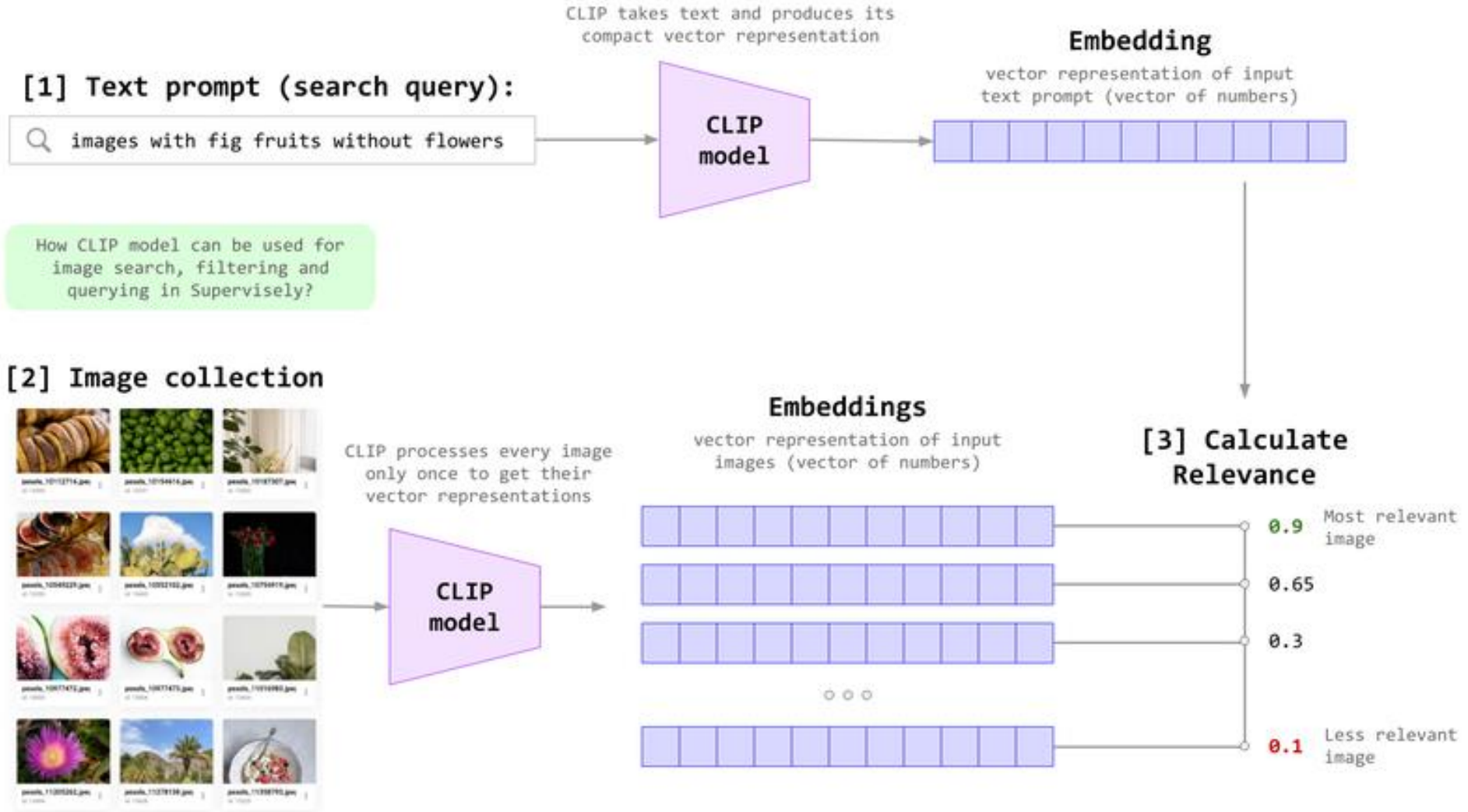


- ✗ a photo of a **broad tailed hummingbird**, a type of bird.
- ✗ a photo of a **calliope hummingbird**, a type of bird.
- ✗ a photo of a **costas hummingbird**, a type of bird.
- ✓ a photo of a **black chinned hummingbird**, a type of bird.
- ✗ a photo of a **annas hummingbird**, a type of bird.

Applications: Image Retrieval

By converting both text queries and images into a common embedding space, CLIP can efficiently retrieve images that are semantically related to textual queries, enhancing search functionalities in databases and online.

Applications: Image Retrieval



Applications: Image Caption

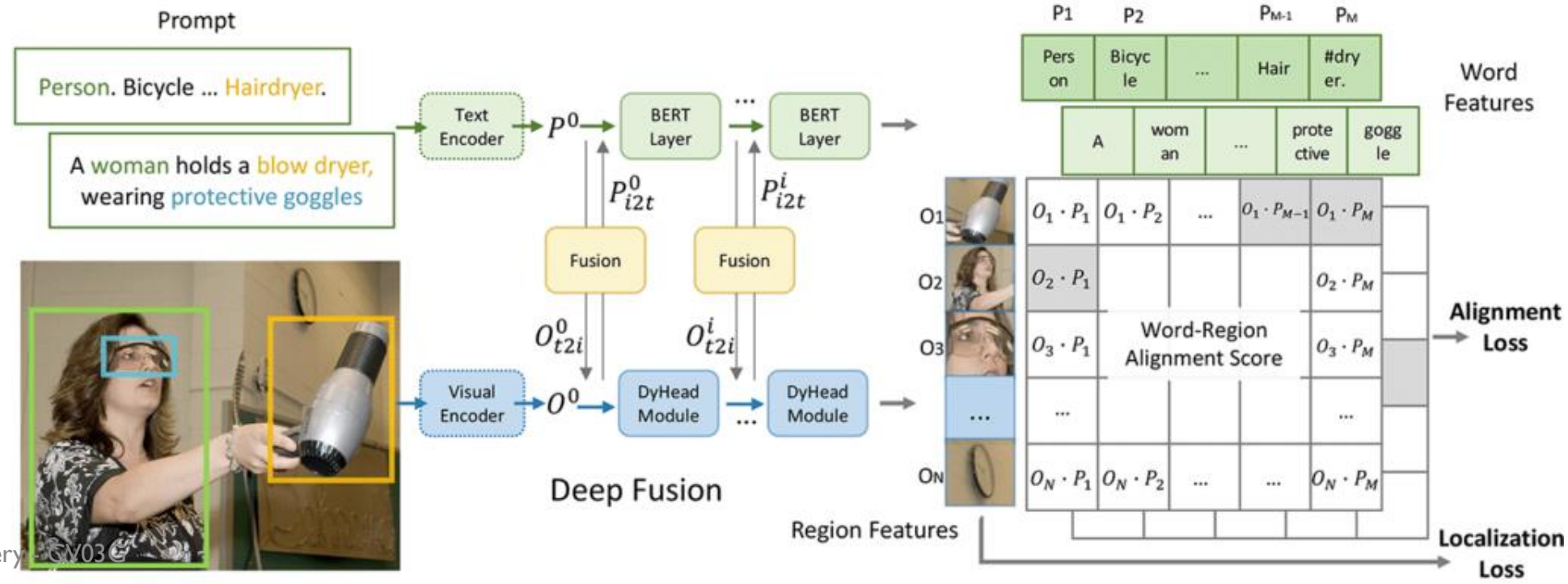
CLIP can assist in generating descriptive captions for images or even help generate images from textual descriptions by linking the textual content with visual representations.



Two people relax at a beach under a tree, enjoying a serene ocean view.

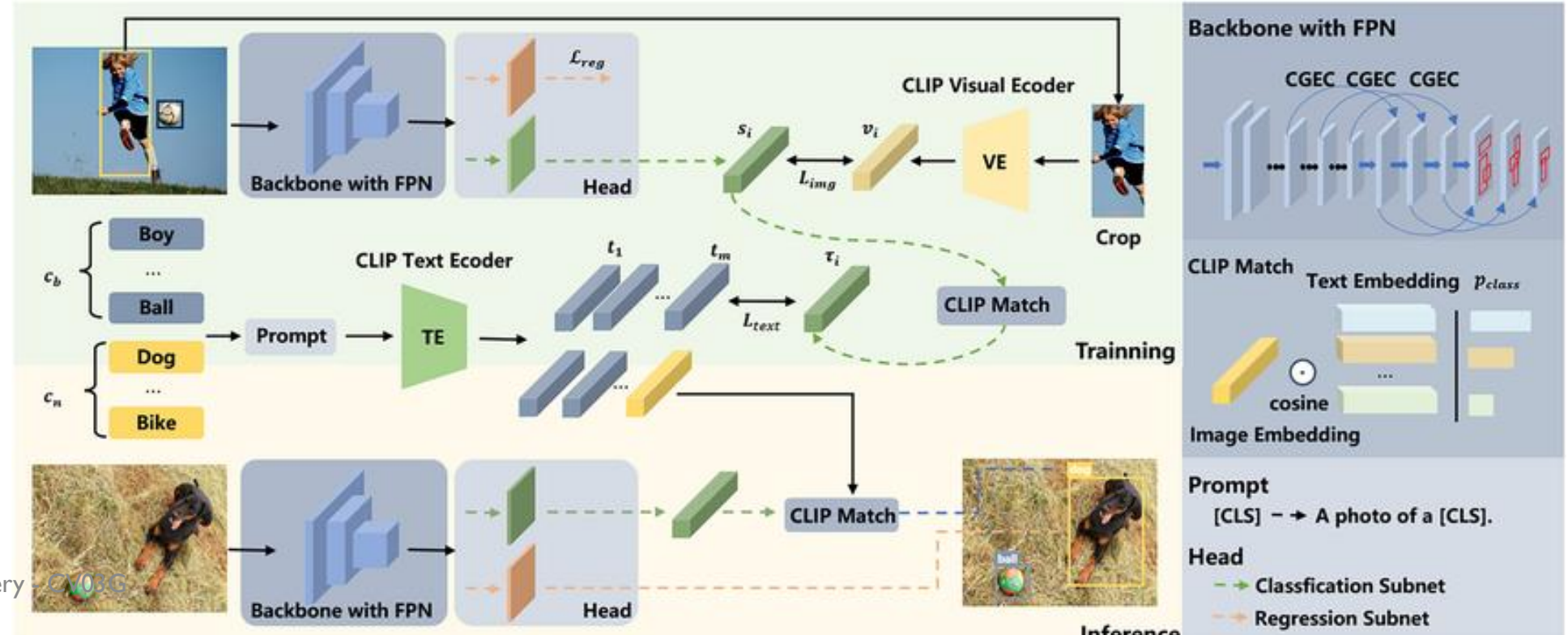
Applications: Object Detection

Although not primarily designed for this, CLIP can be adapted for object detection tasks where objects are detected based on their descriptions rather than traditional bounding box annotations.



Applications: Object Detection

Although not primarily designed for this, CLIP can be adapted for object detection tasks where objects are detected based on their descriptions rather than traditional bounding box annotations.



Applications: Image Generation

Although not primarily designed for this, CLIP can be adapted for image generation.

